C1 Atomic Structure – Knowledge Organiser

AQA A-level Chemistry • Set 1: Subatomic Particles Paper 1

#	Question	Answer
1	Relative mass of a proton	1
2	Relative charge of a proton	+1
3	Relative mass of a neutron	1
4	Relative charge of a neutron	0
5	Relative mass of an electron	1/1840 (≈ 0)
6	Relative charge of an electron	−1
7	Atomic number (*Z*)	Number of protons
8	Mass number (*A*)	Number of protons + neutrons
9	Isotopes	Same number of protons, different number of neutrons
10	Location of most of an atom's mass	Nucleus

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AQA A-level Chemistry • Set 2: Relative Masses and Isotopes **Paper 1**

#	Question	Answer
1	Relative atomic mass (A_r)	Weighted mean mass of an atom ÷ 1/12 mass of a ^{12}C atom
2	Relative isotopic mass	Mass of an atom of an isotope ÷ 1/12 mass of a ^{12}C atom
3	Relative molecular mass (M_r)	Mean mass of a molecule ÷ 1/12 mass of a ^{12}C atom
4	Standard used for relative masses	Carbon-12
5	Why isotopes have the same chemical properties	Same electron configuration
6	Property that differs between isotopes	Mass (density)
7	Type of mass spectrometer in the AQA spec	Time of flight (TOF)
8	Two ionisation methods in TOF	Electrospray and electron impact
9	Ion formed by electrospray ionisation of X	XH^+
10	Ion formed by electron impact ionisation of X	X^+

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AQA A-level Chemistry • Set 3: Mass Spectrometry and Orbitals **Paper 1**

#	Question	Answer
1	Kinetic energy equation	KE = ½mv^2
2	Equation linking velocity, distance and time	$v = d \div t$
3	x-axis of a mass spectrum	m/z
4	y-axis of a mass spectrum	Relative abundance
5	Ions that reach the TOF detector first	Lightest (lowest m/z)
6	Charge of ions detected in TOF	Positive (1+)
7	Maximum electrons in an s subshell	2
8	Maximum electrons in a p subshell	6
9	Maximum electrons in a d subshell	10
10	Maximum electrons in one orbital	2 (opposite spins)

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AQA A-level Chemistry • Set 4: Electron Configuration **Paper 1**

#	Question	Answer
1	Shape of an s orbital	Spherical
2	Shape of a p orbital	Dumbbell
3	Number of orbitals in a p subshell	3
4	Number of orbitals in a d subshell	5
5	Maximum electrons in the third shell	18
6	Subshell filled first: 4s or 3d	4s
7	Subshell emptied first when ions form: 4s or 3d	4s
8	Two Period 4 exceptions to the filling order	Cr and Cu
9	Electron configuration of Cr	[Ar] $3d^5$ $4s^1$
10	Electron configuration of Cu	[Ar] $3d^{10}$ $4s^1$

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AQA A-level Chemistry • Set 5: Ionisation Energy **Paper 1**

#	Question	Answer
1	First ionisation energy	Energy to remove 1 mol of electrons from 1 mol of gaseous atoms
2	Equation for the first ionisation energy of Na	$Na(g) \rightarrow Na^{+}(g) + e^{-}$
3	Second ionisation energy	Energy to remove 1 mol of electrons from 1 mol of gaseous 1+ ions
4	Units of ionisation energy	$kJ\ mol^{-1}$
5	Trend in first IE down a group	Decreases
6	General trend in first IE across a period	Increases
7	Period 3 element with lower first IE than Mg	Al
8	Why Al has a lower first IE than Mg	Outer electron in higher-energy 3p subshell
9	Period 3 element with lower first IE than P	S
10	Why S has a lower first IE than P	Paired 3p electrons repel

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AQA A-level Chemistry • Set 6: Factors, Blocks and Ions **Paper 1**

#	Question	Answer
1	Three factors affecting ionisation energy	Nuclear charge, atomic radius, shielding
2	Evidence for electron shells	Large jumps in successive IEs
3	Evidence for subshells	Drops in first IE at Al and S
4	Block containing Ca	s
5	Block containing Fe	d
6	Block containing Br	p
7	Electron configuration of Fe^{2+}	[Ar] $3d^6$
8	Electron configuration of Fe^{3+}	[Ar] $3d^5$
9	Electron configuration of O^{2-}	$1s^2\ 2s^2\ 2p^6$
10	Isoelectronic	Same electron configuration